

How Do Space and Society Shape Child Neurodevelopment? Interpretable Machine-Learning Evidence from Shanghai

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Abstract

Preschooler neurodevelopment is shaped not only by biological and family conditions but also by everyday urban environments where preschoolers and their caregivers engage in activities. Yet translating public health evidence into precise urban interventions remains challenging as an integrated framework grounded in an actionable spatial representation is still lacking. Existing studies often foreground individual-level associations or isolated environmental attributes, limiting both the fine-grained identification of associations between preschooler neurodevelopment and multidimensional urban environmental configurations and the formulation of spatially targeted planning interventions. This study proposes a spatially explicit framework for planning intervention and applies it to survey data from 12,997 preschoolers in Songjiang, Shanghai, across five neurodevelopmental domains. Results showed that associations between preschooler neurodevelopment and the built environment became more spatially legible after spatial aggregation, with streets providing the clearest operational representation. At the street level, the neurodevelopmental variation exhibited significant spatial structure after accounting for BDC and SES, which could further be explained by built-environment conditions through residual modelling. SHAP analysis showed that place-based urban conditions were relevant across multiple neurodevelopmental domains, whereas human-focused streetscape perceptions exhibited more domain-specific associations. Partial dependence profiles were then used to derive cross-domain lower- and higher-risk environmental ranges, which were integrated into health-oriented street typologies representing distinct built-environment profiles and graded levels of neurodevelopmental risk. High-risk profiles were more concentrated in streets with lower aggregated maternal education. The framework therefore translates individual neurodevelopmental evidence into fine-grained spatial priorities for differentiated and equity-sensitive planning intervention.

Keywords

Preschooler neurodevelopment; Built environment; Socioeconomic differentiation; Spatial inequality; Interpretable machine learning

1. Introduction

Early childhood is a formative stage of human development, during which rapid physiological and neurological changes establish foundations for later learning, behaviour, social participation and wellbeing (Black et al., 2017; Shonkoff et al., 2012). Neurodevelopment during this period is reflected in emerging capacities in communication, motor coordination, problem solving and social interaction. Although these capacities are strongly shaped by biodemographic and household socioeconomic conditions, they also develop through children's continuing interactions with everyday social and physical environments (Bradley & Corwyn, 2002; Minh et al., 2017). Built environment structures opportunities for movement, play, exploration and social contact while also shaping exposure to environmental stressors and unevenly distributed urban resources. Understanding preschooler neurodevelopment in relation to urban environments is therefore relevant not only to public health but also to urban planning, because many environmental conditions are spatially identifiable and potentially modifiable through planning and design.

A growing body of research has associated child health and development with neighbourhood greenness, parks and play spaces, walkability, transport environments, air pollution, service accessibility and, more recently, human-scale streetscape qualities (Christian et al., 2015; Ortegon-Sanchez et al., 2021). Greenness, accessible parks and walkable environments may support cognition, attention, physical activity and social interaction, whereas traffic-related pollution and other environmental stressors have been associated with behavioural difficulties and delayed psychomotor development (Ahmed et al., 2022; Rong et al., 2025). Emerging studies using street-view imagery and virtual audits further capture human-scale qualities such as safety, playability, enclosure and visual complexity that conventional neighbourhood indicators may overlook (McCoy et al., 2022; Shen et al., 2021). Yet evidence remains fragmented across environmental attributes and developmental domains, providing limited understanding of how multidimensional urban conditions jointly constitute environments that are more or less supportive of early development.

This limitation is partly methodological. Most studies estimate adjusted associations using linear, generalised linear or multilevel regression models, often treating environmental attributes as separate exposures (Ahmed et al., 2022; Bell et al., 2020). Such approaches are useful for estimating average associations but may overlook nonlinearities and interactions among correlated environmental conditions. More recent machine-learning approaches can accommodate high-dimensional predictors and complex response structures, while explainable artificial intelligence tools such as SHAP and partial dependence profiles can reveal feature contributions and nonlinear response patterns (England-Mason et al., 2025). However, when findings are reported mainly as predictive performance, feature rankings or predictor-specific curves, they remain difficult to translate into spatially explicit planning knowledge about which environmental conditions and locations should be prioritised for intervention.

A further challenge concerns spatial representation. Evidence on associations between child development and the built environment depends not only on the variables and models used but also on how children and environmental conditions are spatially linked. Spatial aggregation changes the statistical support of an analysis, and estimated associations and geographical patterns may vary with the scale and configuration of spatial units (Openshaw, 1983). Fine-grained units preserve local heterogeneity but may yield unstable estimates when observations are sparse, whereas broader units improve stability while smoothing local differences (Li et al., 2020). Most child health studies nevertheless retain individual children as the primary analytical units, assigning environmental exposures through residential addresses, buffers or network-based service areas (Ha et al., 2019; Villanueva et al., 2023). Although appropriate for estimating individual exposure–outcome associations, these approaches do not directly identify the spatial objects through which planning intervention should occur. Conversely, area-level studies can make geographical inequalities more visible but often conceal variation among finer-grained settings such as streets, plots and public spaces (Christian et al., 2017; Collyer et al., 2022). Existing research has therefore paid limited attention to spatial representations that simultaneously preserve local heterogeneity, provide sufficient statistical stability and correspond to modifiable urban environments. Identifying such a representation is essential for translating child health evidence into targeted planning action.

This planning challenge is also closely connected to socioeconomic differentiation in early childhood development. Parental education, household income and related family resources shape access to responsive care, learning opportunities and supportive material conditions (Bradley & Corwyn, 2002; Walker et al., 2011). These inequalities are spatially reproduced through unequal exposure to urban resources and environmental burdens. Socioeconomically disadvantaged children may

experience greater exposure to air pollution and more limited access to high-quality green space and supportive neighbourhood environments (Clark et al., 2014; Williams et al., 2020). Moreover, family resources and neighbourhood contexts may influence how environmental opportunities are accessed and used, meaning that associations between built-environment conditions and neurodevelopment may vary across socioeconomic contexts (Minh et al., 2017). Yet inequalities in environmental exposure and socioeconomic variation in environmental association patterns are rarely examined within the same analytical framework.

To address these gaps, this study develops a spatially explicit explainable artificial intelligence framework for translating evidence on preschooler neurodevelopment into interpretable and actionable spatial knowledge. Drawing on survey data from 12,997 children aged 3–5.5 years in Songjiang, Shanghai, the study first compares alternative spatial representations to identify the unit that best balances statistical stability, local differentiation and planning operability. At the selected spatial unit, multidimensional built-environment conditions are examined in relation to five domains of preschooler neurodevelopment, incorporating both place-based urban conditions and human-focused streetscape characteristics. SHAP is used to identify influential environmental features, while partial dependence profiles are used to characterise nonlinear lower- and higher-risk environmental ranges. These ranges are then integrated into health-oriented street typologies representing different environmental configurations and levels of modelled neurodevelopmental risk. Finally, the distribution of these typologies and associated environmental resources is compared across streets stratified by aggregated maternal education levels to examine socioeconomic differences in access to supportive urban environments.

Accordingly, this study addresses the following research questions: (1) Which spatial representation most clearly reveals associations between preschooler neurodevelopment and the built environment while remaining operational for planning intervention? (2) Which place-based urban conditions and human-focused streetscape characteristics are most strongly associated with preschooler neurodevelopment, and what nonlinear environmental ranges can be identified? (3) How are health-oriented street typologies and supportive environmental resources distributed across different socioeconomic contexts?

By jointly addressing spatial representation, multidimensional environmental interpretation and socioeconomic differentiation in resource distribution, the study moves beyond predictor-specific associations towards a finer-grained spatial framework for identifying where lower- and higher-risk environments are located and which urban conditions may warrant planning attention.

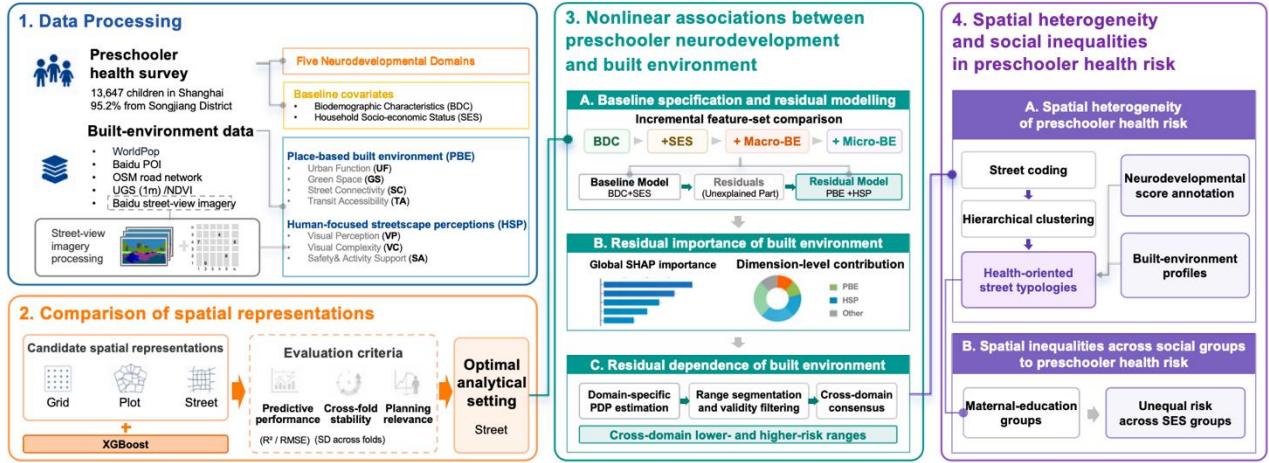


Figure 1 Research flow: reclassifying urban space based on heterogeneous preschooler neurodevelopment-environment nonlinear patterns

2. Method

2.1. Reclassifying urban space with nonlinear patterns between preschooler neurodevelopment and built environment

This study develops a spatially explicit framework to translate evidence on preschooler neurodevelopment into actionable knowledge for urban intervention (Figure 1). The framework first compares alternative spatial representations, including grids, plots and street segments, to identify the analytical unit at which associations between preschooler neurodevelopment and the built environment become most spatially legible and operational for planning. Residual modelling is then applied to examine built-environment signals remaining beyond biodemographic characteristics and household socioeconomic conditions. SHAP analysis identifies influential environmental attributes, while partial dependence profiles characterise nonlinear associations and derive lower- and higher-risk environmental ranges. These cross-domain ranges are further translated into street-level environmental profiles and clustered to construct health-oriented street typologies representing distinct built-environment configurations and graded levels of modelled neurodevelopmental risk. Finally, the framework examines the spatial distribution and socioeconomic differentiation of these typologies, identifying where developmental vulnerability overlaps with modifiable environmental constraints. The framework therefore provides a pathway from individual neurodevelopmental evidence to spatially targeted planning intervention.

2.2. Data processing

Preschooler neurodevelopment is associated with a combination of biodemographic characteristics and household socioeconomic conditions, as well as the built environments in which everyday activities occur (Black et al., 2017; Sania et al., 2019). Accordingly, this study

organised the explanatory variables into three domains: biodemographic characteristics (BDC), household socioeconomic conditions (SES) and built-environment conditions (BE). BDC and SES represented established non-environmental covariates, whereas BE captured potentially modifiable urban conditions relevant to preschoolers' everyday activities and experiences. BE was further characterised from two complementary perspectives: Place-based built-environment conditions (PBE), representing the provision, accessibility and spatial configuration of urban resources, and human-focused streetscape perceptions (HSP), representing the visual and spatial qualities encountered at street.

(1) Biodemographic characteristics (BDC) and household socioeconomic conditions (SES)

Individual-level covariates were grouped into two categories based on prior evidence and data availability: (1) **Bio-demographic characteristics (BDC)** included preschooler sex (male, female), age at assessment, birth weight, birth order, admission to a neonatal intensive care unit (NICU; no, yes), twin or multiple-birth status (no, yes), current body mass index, and handedness preference (right, left, unclear), together with the current height and weight of both parents. (2) **Household socioeconomic status (SES)** included maternal and paternal educational attainment (six ordered levels from illiterate to graduate education), parental occupation (manual or service work, self-employed, professional or managerial employment, unemployed), annual per capita household income (eight ordered categories ranging from $\leq 10,000$ CNY to $>1,000,000$ CNY), household structure (nuclear, stem, joint, single-parent), and parental marital status (first marriage, remarried, separated, divorced or widowed). Continuous variables were retained in their original measurement units, while categorical and ordinal variables were coded according to their original classifications.

(2) Place-based built-environment conditions (PBE)

Place-based built-environment conditions were derived from multisource geospatial data and calculated for each analytical unit and its surrounding neighbourhoods. Four domains were considered: urban function (UF), green space (GS), street connectivity (SC) and transit accessibility (TA). UF included functional diversity, POI intensity and access to educational and healthcare facilities. Educational and healthcare facilities were identified separately from general POIs because of their

direct relevance to preschooler development and healthcare access. GS comprised mean NDVI and the total area of accessible green space, representing vegetation conditions and local green-space availability. SC was measured using angular choice and angular integration, reflecting potential through-movement and network accessibility. TA included intersection density, metro and bus-stop availability, and distances to the nearest metro station and bus stop. Definitions, units and calculation methods are provided in [Table 2](#)

Table 2 Definitions and measurement of Place-based built-environment conditions

Indicator domain	Indicator group	Variable	Unit / Coding	Definition
Urban Function (UF)	Population	Pop%R	Count	Population within radius R
	Land Use Diversity	Div	0–1	Shannon entropy of POI categories
	General POI Intensity	OFF / PUB / REC / CAT / TRA / HOS / RET / CUL / EDU / HOT / PAR	Count	Number of facilities in each POI category within buffer
	Education Accessibility	DistEdu	m	Distance to the nearest educational facility
		DistKinder / DistPrimary / DistMiddle / DistCollege	m	Distance to the nearest educational facility at each school level
		Edu%R/	Count	Number of educational facilities within radius R
		Kinder%R/ Primary%R / Middle%R / College%R	Count	Number of educational facilities at each school level within radius R
	Healthcare Accessibility	DistHosp	m	Distance to nearest hospital
		Hosp%R	Count	Number of hospitals within radius R
	Vegetation	NDVI%R	-1 to 1	Mean Normalised Difference Vegetation Index within radius R
Green Space (GS)	Accessible Green Space	UGS%R	m ²	Total area of accessible green space within radius R, derived from 1 m-resolution land-cover data
Street Connectivity (SC)	Centrality	CHO%R	Value	Angular betweenness within radius R
		INT%R	Value	Angular closeness within radius R
	Network Density	Int_cnt%R	Count	Number of street intersections within radius R
Transit Accessibility (TA)	Transit Supply	Metro_cnt%R	Count	Number of metro stations within radius R
		Bus_cnt%R	Count	Number of bus stops within radius R
	Transit Proximity	Dist_metro	m	Distance to nearest metro station
		Dist_bus	m	Distance to nearest bus stop

Note: Variables with the suffix *R* were calculated at multiple spatial scales. Population, intersection, and transit-supply indicators were calculated within buffer radii of 100, 300, 500, and 1,000 m. Angular choice and integration were calculated using network radii of 500, 1,000, 2,000, 4,000, 6,000, 9,000, 10,000, 15,000, 20,000, 50,000, 100,000, and 150,000 m.

(3) Human-focused streetscape perceptions (HSP)

Preschooler development is associated not only with access to neighbourhood resources, but also with the

qualities of the environments they encounter in everyday life. Street-level visual form and spatial organisation may support or constrain opportunities for play, exploration, social interaction and outdoor mobility, while traffic and safety conditions may influence the willingness of preschoolers and their caregivers to engage with the local environment (Christian et al., 2015; Gemmell et al., 2023). To capture these qualities, which are not adequately represented by place-based measures, this study derived human-focused streetscape indicators (HSP) from street-view imagery.

Each image was first processed using the DeepLabV3+ semantic segmentation architecture (Chen et al., 2018), which assigned every pixel to an urban-scene category, including roads, pavements, buildings, vegetation, sky, motor vehicles and pedestrians. Composition-based indicators were calculated from the proportion of pixels belonging to each category. These indicators quantified the visible presence of individual streetscape elements, such as sky openness, green view, built enclosure, pedestrian space and motor-vehicle presence.

Because pixel proportions describe the amount of each element but not how those elements are spatially arranged, topology-based indicators were additionally derived from semantic adjacency relationships (Zhang et al., 2024). Each segmented image was represented as a categorical matrix, and adjacency events were identified when neighbouring pixels belonged to different semantic classes. These events were aggregated and normalised to construct image-level semantic adjacency matrices, from which selected inter-category relationships were extracted to represent streetscape interfaces relevant to

visual experience, environmental complexity and pedestrian activity.

The resulting HSP variables therefore captured two complementary dimensions of the streetscape: the visible composition of environmental elements and the spatial relationships among them. Based on their relevance to preschoolers' street-level experience and activity, the indicators were organised into three domains: visual perception, visual complexity, and safety and activity support. Detailed definitions and calculation methods are presented in Table 3.

Based on the explanatory variables defined above, the predictive mapping from the measured predictors to each neurodevelopmental outcome was represented using a general modelling framework. For analytical unit i and neurodevelopmental domain d , the relationship was expressed as:

$$\hat{Y}_{i,d} = f_d(\mathbf{X}_i^{BDC}, \mathbf{X}_i^{SES}, \mathbf{X}_i^{BE}), \mathbf{X}_i^{BE} = (X_i^{PBE}, X_i^{HSP}) \quad (1)$$

where $\hat{Y}_{i,d}$ denotes the predicted neurodevelopmental outcome; $\mathbf{X}_i^{BDC}$, $\mathbf{X}_i^{SES}$ and $\mathbf{X}_i^{BE}$ denote the measured biodemographic, household socioeconomic and built-environment predictors, respectively. $\mathbf{X}_i^{BE}$ comprises place-based built-environment conditions (X_i^{PBE}) and human-focused streetscape characteristics (X_i^{HSP}). $f_d(\cdot)$ denotes the fitted prediction function for neurodevelopmental domain d , which may capture nonlinearities and interactions among predictors depending on the selected algorithm.

Table 3 Definitions and measurement of human-focused streetscape perception indicators

Indicator domain	Indicator group	Variable	Unit / Coding	Definition
Visual Perception	Openness	Sky Openness	%	Proportion of sky pixels.
	Natural exposure	Green View Index	%	Proportion of vegetation pixels.
	Built enclosure	Built Enclosure Ratio	%	Proportion of building, wall, and fence pixels, representing visible built enclosure.
	Traffic visibility	Vehicular Disturbance	%	Proportion of car, truck, bus, and motorcycle pixels, representing visible motor-vehicle presence.
Visual Complexity	Compositional diversity	Visual Entropy	value	Shannon entropy calculated from the distribution of semantic pixel categories.
	Adjacency density	Visual Crowdedness	value	Average degree of the semantic adjacency matrix, representing the density of visual element adjacency.
	Skyline interface	Skyline Articulation	value	Normalised adjacency weight between sky and building pixels, representing the complexity of the skyline interface.
Safety and Activity Support	Nature contact	Nature Contact Proximity	value	Normalised adjacency weight between sidewalk/person and vegetation pixels, representing street-level contact with greenery.
	Pedestrian space	Walkable Surface Ratio	%	Proportion of sidewalk pixels, representing visible pedestrian space.
	Physical separation	Pedestrian–Vehicle Separation	value	Normalised adjacency weight between sidewalk and fence/wall pixels, representing physical separation along pedestrian space.
	Traffic exposure	Roadside Exposure Risk	value	Normalised adjacency weight between sidewalk and road/vehicle pixels, representing pedestrian exposure to traffic elements.
	Dynamic conflict	Dynamic Conflict	value	Normalised adjacency weight between sidewalk and

	Sources	rider/bicycle pixels, representing potential non-motorised movement conflicts.
Street interface	Street Surveillance	Normalised adjacency weight between building and sidewalk pixels, representing the potential interface between building frontage and pedestrian space.
	value	

2.3. Comparison of spatial representations

Spatial representations were constructed using four grid resolutions (250 m, 500 m, 750 m and 1 km), together with plot and street segments. To ensure a fair comparison, all spatial representations were evaluated using the same predictor sets, modelling algorithm and cross-validation framework. They were then compared in terms of out-of-sample predictive performance, cross-validation stability, sample retention and planning relevance, with the preferred representation selected by jointly considering statistical performance and operational applicability.

2.4. Nonlinear associations between preschooler health and built environment

2.4.1. Baseline specification and residual modelling

Previous multilevel and neighbourhood-scale machine-learning studies have generally found that contextual and built-environment conditions retain detectable associations with preschooler neurodevelopment, but contribute less explanatory or predictive value than sociodemographic and household socioeconomic factors. To examine whether a similar pattern was observed in the present study, models with progressively expanded feature sets were first compared across the five neurodevelopmental domains. Five specifications were compared: BDC; BDC + SES; BDC + SES + PBE; the full model comprising BDC + SES + PBE + HSP; and an environment-only model comprising PBE + HSP. All specifications were evaluated using the same cross-validation framework. Changes in R^2 , RMSE and MAE were used to assess the additional out-of-sample predictive value associated with household socioeconomic and built-environment variables.

The incremental comparison quantified the additional predictive value of built-environment conditions, but did not directly characterise how environmental conditions were associated with neurodevelopmental variation beyond that predicted by biodemographic and household socioeconomic characteristics. A residual modelling strategy was therefore used to provide a more focused environmental interpretation. The BDC + SES specification was adopted as the non-environmental benchmark. For analytical unit i and neurodevelopmental domain d , its out-of-fold prediction was expressed as: $\hat{Y}_{i,d} = f_d^{base}(X_i^{BDC}, X_i^{SES})$. The corresponding residual was calculated as: $\hat{r}_{i,d} = Y_{i,d} - \hat{Y}_{i,d}$, where $Y_{i,d}$ denotes the observed neurodevelopmental outcome and $r_{i,d}$ represents the variation not accounted for by biodemographic characteristics and household socioeconomic conditions.

The built-environment residual model was then specified as: $\hat{r}_{i,d} = g_d(X_i^{BE}) + \varepsilon_{i,d}$, where $g_d(\cdot)$ represents the modelled association between built-environment conditions and the remaining neurodevelopmental variation. The resulting residual predictions were subsequently used for SHAP and partial-dependence analyses to identify potentially modifiable environmental characteristics and planning-relevant environmental ranges.

2.4.2. Residual importance of built environment with SHAP

For each neurodevelopmental domain d , SHAP analysis was applied to the fitted built-environment residual model to quantify the contribution of each environmental variable to the predicted residual variation. Global predictor importance was quantified using the mean absolute SHAP value. Importance patterns were then compared across place-based and human-focused environmental dimensions and across developmental domains to distinguish predictors that were consistently prominent from those showing domain-specific importance.

2.4.3. Residual dependence of built environment with PDP

For the influential built-environment features identified through SHAP, partial dependence profiles (PDP) were derived from the fitted built-environment residual models to characterise their nonlinear associations with the predicted residual variation. For feature j and neurodevelopmental domain d , the partial dependence function was defined as:

$$PD_{j,d}(z) = \frac{1}{n} \sum_{i=1}^n g_d(z, X_{i,-j}^{BE}) \quad (2)$$

where $g_d(\cdot)$ denotes the fitted residual model, z is a specified value of feature j , and $X_{i,-j}^{BE}$ represents the remaining built-environment predictors for analytical unit i . The function therefore represents the average predicted residual while varying the target feature and averaging over the empirical distribution of the remaining predictors. To enable comparison across features and neurodevelopmental domains, each PDP curve was mean-centred relative to its average predicted residual value. The centred profiles were used to identify nonlinear response patterns, including potential thresholds, plateaus and changes in response direction.

A cross-domain consensus framework was developed to identify built-environment ranges consistently associated with lower or higher neurodevelopmental risk across the five domains. Each mean-centred PDP curve was

partitioned into contiguous positive and negative segments relative to the zero baseline. The boundaries, width, direction and maximum response magnitude of each segment were recorded. To exclude unstable or weakly supported patterns, an interval was retained only when it contained at least 5% of observations, covered at least 5% of the feature's q_{05} – q_{95} range and reached at least 40% of the total response amplitude of the corresponding profile.

The retained intervals were then projected onto a common feature-value axis and compared across the five neurodevelopmental domains. A cross-domain consensus range was identified when at least three domains exhibited responses in the same direction over the same interval. As higher neurodevelopmental scores indicated more advanced development, positive PDP responses were interpreted as lower-risk ranges, whereas negative PDP responses were interpreted as higher-risk ranges. Adjacent intervals with the same response direction were merged when the intervening gap did not exceed 1% of the feature's 5th–95th percentile range. The resulting consensus ranges were subsequently used to recode street-level built-environment characteristics and construct health-oriented street typologies representing distinct environmental profiles and graded levels of modelled neurodevelopmental risk.

2.5. Spatial heterogeneity and social inequalities in preschooler health risk

2.5.1. Spatial heterogeneity of preschooler health risk

The cross-domain consensus analysis identified lower- and higher-risk ranges for each built-environment feature. A further procedure was therefore required to determine whether the observed feature value of each street segment fell within one of these ranges and, when it fell outside all ranges, how far it was from the nearest range. Range–Distance Decay Coding (RDDC) scheme was developed for this purpose. For built-environment feature j , the observed value x_{ij} for street segment i was assigned a code of $+1$ when it fell within a lower-risk consensus range and -1 when it fell within a higher-risk consensus range. When x_{ij} fell outside all identified ranges, its code decayed towards zero according to its distance from the nearest consensus-range boundary:

$$\text{code}_{ij} = \begin{cases} 1, x_{ij} \in L_j \\ -1, x_{ij} \in H_j \\ \sigma_{jk} \exp\left(-\frac{d_{ijk}}{w_{jk}}\right), x_{ij} \notin L_j \cup H_j \end{cases} \quad (3)$$

where L_j and H_j denote the sets of lower- and higher-risk consensus ranges for feature j , respectively. For values outside all consensus ranges, k denotes the nearest range, σ_{jk} equals $+1$ when that range is lower-risk and -1 when it is higher-risk, d_{ijk} is the distance from x_{ij} to

the nearest boundary of range k , and w_{jk} is a decay bandwidth scaled to the observed range of feature j . The coding therefore preserves the signal of the nearest cross-domain association while allowing the strength of that association to decrease continuously with distance from the identified range.

Each street was subsequently represented by a multidimensional RDDC vector containing its coded values across the selected built-environment features. Hierarchical clustering was applied to these vectors to identify street segments with similar cross-domain built-environment association profiles. The resulting clusters were interpreted using the original feature values, RDDC scores and aggregated scores across the five neurodevelopmental domains. Neurodevelopmental scores were excluded from clustering and used only to characterise the resulting clusters. Based on their environmental profiles and graded levels of neurodevelopmental risk, the clusters were assigned health-oriented street-typology labels for subsequent analyses of spatial heterogeneity and socioeconomic differentiation.

2.5.2. Spatial inequalities across social groups to preschooler health risk

Following the construction of the health-oriented street typologies, socioeconomic differentiation was examined by dividing street segments into three strata based on aggregated maternal education. Cross-tabulations and stratum-specific proportions of the street typologies were used to examine whether lower- and higher-risk typologies were unevenly distributed across street-level education strata.

3. Study area and data

3.1. Study area: Songjiang town in Shanghai

Songjiang town is located in south-western Shanghai and contains Songjiang New City, one of the municipality's five designated new cities (Figure 2a). Its development combines a long-established urban settlement with successive phases of satellite-town, university-town and new-city expansion. This trajectory has produced a heterogeneous urban structure comprising mature built-up areas, comprehensively planned residential districts and areas undergoing continued urban–peripheral transition. The coexistence of established and newly developed environments provides a suitable setting for examining fine-grained spatial variation in preschooler neurodevelopment and its associations with built-environment conditions.

The available preschooler health data were also highly concentrated in Songjiang (Figure 2b). This concentration primarily reflected differences in data availability rather than the underlying spatial distribution of preschoolers. Records from Songjiang were available relatively comprehensively through its district-level maternal and child health service system, whereas

comparable individual-level data from other districts were constrained by confidentiality and data-governance requirements. Given that fine-scale spatial aggregation

requires a sufficiently large and spatially continuous sample, Songjiang was selected as the empirical study area.

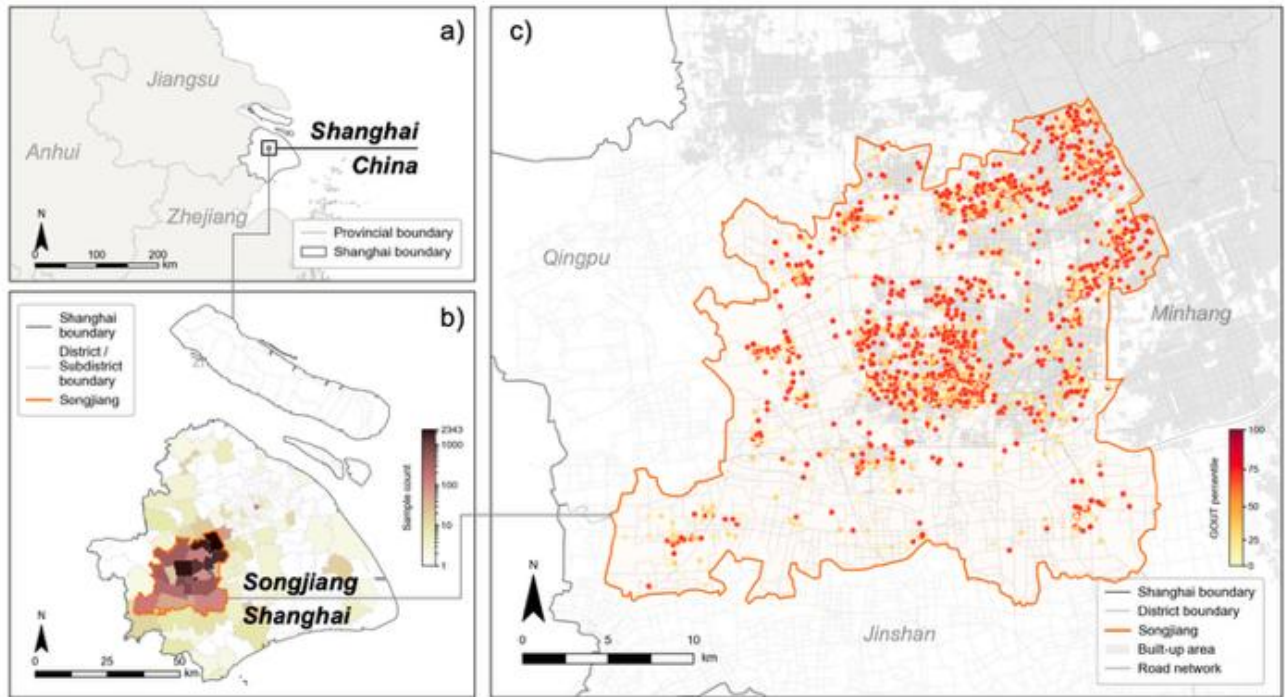


Figure 2 Study area and spatial distributions of the sample. (a) Location of Shanghai in eastern China; (b) location of Songjiang District within Shanghai and the subdistrict-level distribution of sample counts; and (c) spatial distribution of neurodevelopmental outcomes in Songjiang District, with GOUT percentile scores shown as an example.

3.2. Preschooler neurodevelopment data (Y) and covariates (X^{BDC} , X^{SES})

Preschooler neurodevelopment data were drawn from a nationwide survey coordinated through a maternal and child health institution in Shanghai targeting children aged 3–5.5 years. Their neurodevelopment was assessed using the Ages & Stages Questionnaires, Third Edition (ASQ-3), covering communication (GOUT), gross motor (CDDZ), fine motor (JXDZ), problem-solving (JJWT) and personal-social development (GRSH). Higher scores indicate more advanced development in the corresponding domain. The final Shanghai sample comprised 13,647 preschoolers, of whom 12,997 (95.2%) resided in Songjiang and formed the analytical sample. Within Songjiang, the geocoded observations covered the principal built-up areas, with higher concentrations around major residential centres and lower densities in peripheral areas (Figure 2c).

Biodemographic characteristics (BDC) and household socioeconomic conditions (SES) used as explanatory variables were collected through the same questionnaire, with their detailed definitions and coding described in section 3.2.

3.3. Spatial representation and sample aggregation

Spatial representation determines how individual health observations are aggregated and interpreted within urban space. Six candidate spatial representations were constructed in this study: regular grids of 250, 500, 750 and 1,000 m, plots and street segments. Geocoded observations were assigned directly to the grid and plot units containing them. As street segments are linear features without areal extent, individual observations were linked to street segments using a Gaussian soft-assignment approach. Each child observation was spatially allocated to nearby street segments within a 300 m radius, with closer segments receiving greater weights according to the Gaussian distance-decay function. The aggregated neurodevelopmental scores for each street segment were then calculated based on the weighted contributions of linked observations. A 300 m assignment radius was selected because it approximated the local scale of Songjiang's street network, where cleaned street segments had a mean length of 335.33 m and a median length of 287.13 m. A minimum threshold of five linked observations was selected because it retained sufficient sample coverage while reducing uncertainty in the aggregated outcomes for each spatial unit. This threshold was then applied consistently across all six candidate spatial representations, with units containing fewer than five linked observations excluded from subsequent analysis. Figure 3a–c illustrates the

retained units and preschooler observations for the 500 m grid, plot and street-segment representations. After filtering, the grid and street representations retained more than 96% of the original observations, whereas the plot representation retained only 73.72% because of its

more fragmented spatial coverage. Sample retention was therefore included, alongside predictive performance, and planning operability in the subsequent evaluation of candidate spatial representations.

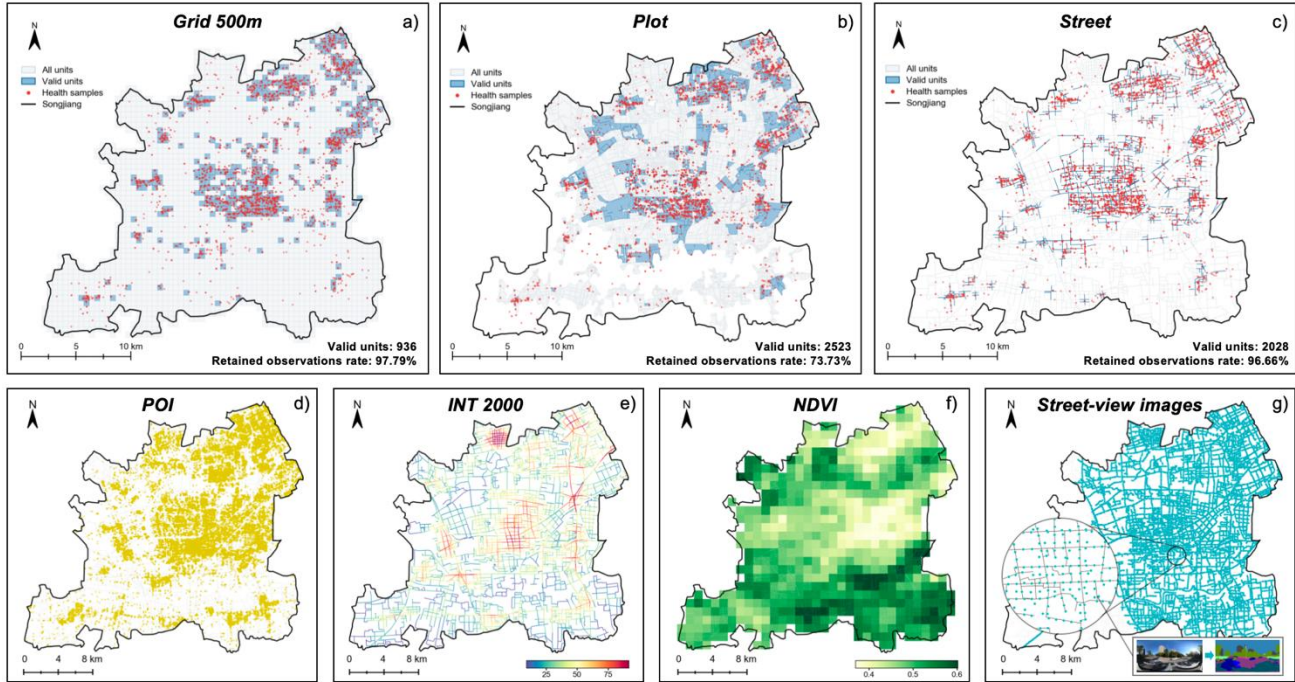


Figure 3 **Candidate spatial representations and built-environment data in Songjiang District.** (a-c) Observations assigned to 500 m grid cells, plot units and street segments; (d) spatial distribution of points of interest; (e) angular integration within a 2,000 m network radius; (f) Normalised Difference Vegetation Index; and (g) spatial coverage of street-view images.

3.4. Built-environment data (X^{BE})

Multisource spatial data were assembled to characterize place-based built-environment conditions and human-focused streetscape perceptions. Points of interest (POIs) were obtained from the Amap Open Platform for 2025 and classified into major urban-function categories, with educational, healthcare and public-transport facilities retained as separate categories. Street-network data were obtained from OpenStreetMap and processed following Liu and Jiang (2012), including road-type filtering, centreline extraction and network segmentation. The processed network was used to construct the segment-based street network model and calculate multiscale indicators of street connectivity and accessibility, including integration, choice and intersection counts.

Vegetation conditions were represented by the annual maximum NDVI for Shanghai in 2021, derived from the 1 km MOD13A3 monthly vegetation-index product (Didan, 2021). Fine-grained green-space availability was measured using UGS-1m, a 1 m-resolution urban green-space dataset covering 31 major Chinese cities (Shi et al., 2023). These datasets captured vegetation conditions at complementary spatial resolutions.

Street-view sampling points were generated at 50 m intervals along road centrelines, and 360° static panoramic images were retrieved through the Baidu Map

Open Platform. The images were used to derive the composition- and topology-based human-focused streetscape perceptions described in the section 3.2. Together, these datasets captured complementary dimensions of urban built-environment conditions. Selected data are illustrated in Figure 3.

4. Empirical Result

4.1. Street segments best reveal associations between preschooler neurodevelopment and built environment

The analysis first examined which alternative spatial representation could provide a legible and operational basis for modelling. The six candidate representations were evaluated using XGBoost, and were compared with predictive performance, sample retention and planning relevance considered jointly.

As shown in Figure 4, street- and plot-based models generally outperformed the grid-based alternatives. Street-based models achieved the highest mean out-of-fold R^2 of 0.480 and maintained comparatively low RMSE values across the five neurodevelopmental domains. Plot-based models produced the lowest mean RMSE and the second-highest mean R^2 of 0.338. Performance among the grid-based models was weaker and generally declined as cell size increased. Mean R^2

decreased from 0.274 for the 250 m grid to approximately 0.133 for the 500 m and 750 m grids, and

further to 0.014 for the 1 km grid, for which some validation folds produced negative R^2 values.

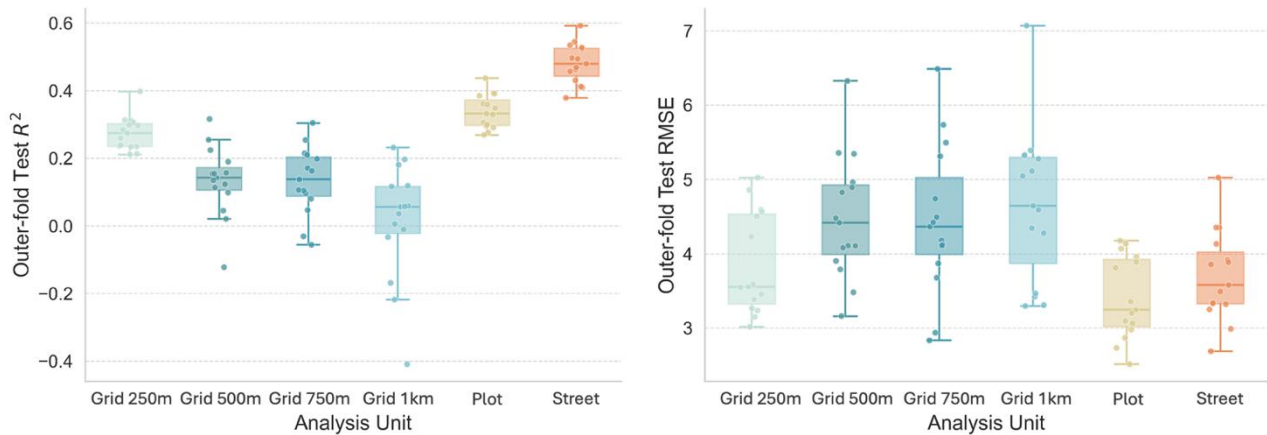


Figure 4 Out-of-fold predictive performance across alternative spatial units. The left panel shows the distribution of out-of-fold R^2 values, and the right panel shows the corresponding RMSE values, across the five neurodevelopmental domains and outer validation folds. Points represent individual fold-level estimates.

Sample retention further distinguished the two best-performing representations. After spatial aggregation and application of the minimum-count threshold, street segments retained 96.66% of eligible observations, compared with 73.72% for plots (Figure 3), whose more fragmented spatial coverage resulted in greater sample loss. Street segments also provided a fine-grained and spatially continuous representation of shared streetscape environments that can be directly modified through planning and design. They were therefore selected as the analytical unit for subsequent modelling and interpretation.

4.2. Residual modelling with built-environment variables for model improvement

The incremental modelling results showed that biodemographic characteristics (BDC) and household socioeconomic conditions (SES) provided the strongest predictive contributions across the five neurodevelopmental domains (Figure 5a). Adding place-based built-environment (PBE) and human-focused streetscape perception (HSP) variables further improved model performance, but with relatively modest gains ($\Delta R^2 = 0.019$ – 0.024 compared with the BDC + SES model). The environment-only model achieved comparable performance for GOUT and JJWT but was less effective for motor-related outcomes and GRSH, suggesting that built-environment variables contained relevant predictive information partly overlapping with individual and household characteristics.

Feature-importance analysis of the full model highlighted the dominant roles of preschooler characteristics, parental socioeconomic conditions and family structure, while environmental factors related to service accessibility, urban functions, green space and streetscape perceptions also contributed across different

developmental domains (Figure 5b–c). However, these rankings reflected the joint predictive structure of all variables and did not isolate the environmental contribution beyond biodemographic and socioeconomic factors.

Therefore, the BDC + SES model was used as the reference specification for residual analysis. Its out-of-fold residuals showed significant spatial autocorrelation across all domains (Moran's $I \approx 0.32$ – 0.40 ; Figure 5d), indicating remaining spatially structured variation. Models incorporating PBE and HSP explained part of this residual variation, achieving positive R^2 values of 0.049 – 0.067 across domains. This residual-based analysis provided a clearer basis for identifying built-environment patterns associated with preschooler neurodevelopment.

4.3. Nonlinear associations between preschooler neurodevelopment and built environment

4.3.1. Residual importance of built environment across neurodevelopmental domains

Based on the residual models, SHAP analysis identified domain-specific patterns in the contribution of built-environment features across the five neurodevelopmental domains (Figure 6). Built-environment contributions were generally stronger for JXDZ and CDDZ, while GOUT, JJWT and GRSH showed lower but distinct attribution patterns.

At the dimension level, place-based urban conditions demonstrated broader associations across domains, whereas human-focused streetscape perceptions showed more domain-specific contributions. Urban functions, street connectivity and transit accessibility were consistently influential, reflecting their role in shaping children's access to destinations, mobility opportunities and daily activities. In contrast, streetscape qualities

related to visual perception, complexity and safety exhibited greater variation across developmental domains, suggesting more context-dependent associations.

Three general environmental profiles were identified. Motor-related domains (JXDZ and CDDZ) were mainly associated with movement-supportive environments characterised by connectivity, accessibility and safer outdoor conditions. JJWT showed stronger associations with cognitively stimulating environments involving

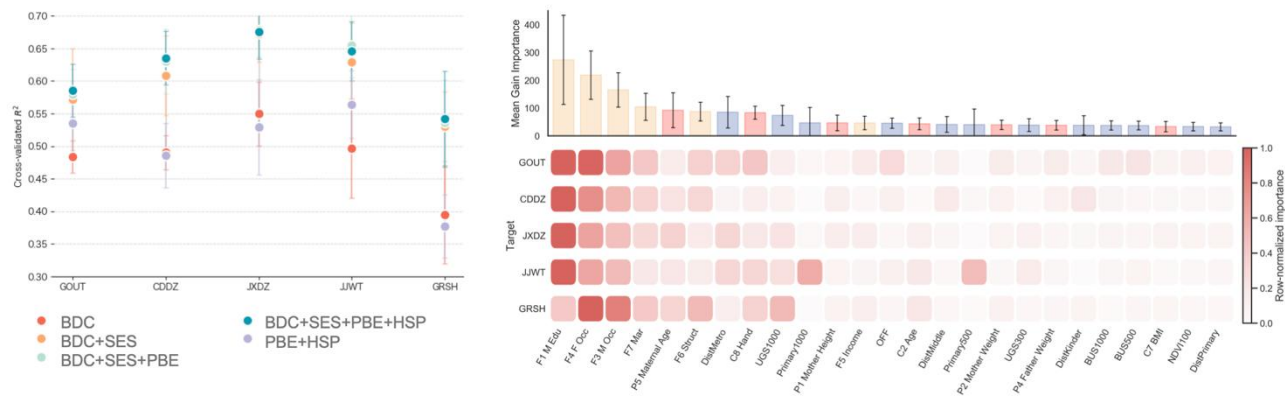


Figure 5 **Incremental predictive performance and feature-importance structures across neurodevelopmental domains.** (a) Cross-validated performance of models incorporating successive feature sets; (b) mean feature importance across the five domains; and (c) domain-specific feature-importance pattern.

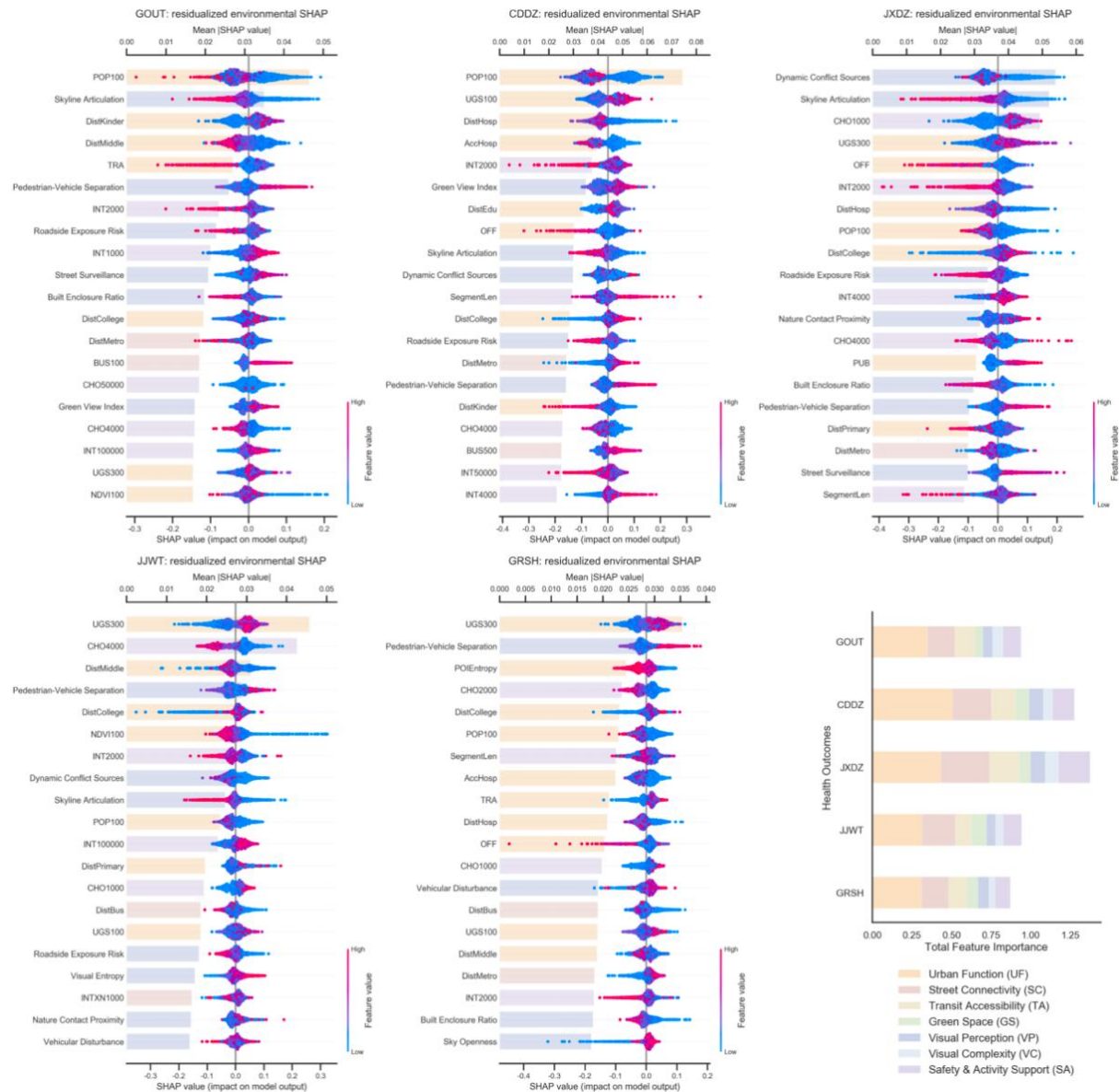


Figure 6 SHAP-based environmental feature importance across the five neurodevelopmental domains. Panels (a–e) present SHAP beeswarm plots for GOUT, CDDZ, JXDZ, JJWT and GRSH. Panel (f) shows the aggregate feature importance of the seven place-based and human-focused environmental dimensions.

visual complexity and natural exposure. GOUT and GRSH were more related to socially supportive environments integrating greenery, functional diversity and pedestrian-oriented spaces.

4.3.2. Residual dependence pattern and cross-domain consensus of built environment

Building on the SHAP results, partial dependence profiles were generated for the 25 highest-ranked environmental features to identify nonlinear response patterns and potential intervention ranges. Across the five neurodevelopmental domains, widespread threshold- and interval-dependent associations were observed. The identified lower- and higher-risk ranges of these features were subsequently used for street-level RDDC coding. Figure 7 presents nine representative features selected based on the clarity, cross-domain

consistency and planning relevance of their response profiles. Three recurring response patterns were identified.

Minimum-threshold patterns were observed when lower-risk responses emerged only after environmental attributes exceeded certain levels. For example, UGS300 was mainly associated with higher-risk responses between approximately 18 and 43, but shifted towards a lower-risk range above about 45. Similarly, Green View Index showed a lower-risk association from approximately 0.13, while Pedestrian–Vehicle Separation became more consistently associated with lower risk above approximately 0.007. These patterns suggest that supportive environmental effects may require sufficient levels of provision rather than simple environmental presence.

Upper-limit patterns occurred when lower-risk responses were observed at low-to-moderate levels but shifted towards higher-risk responses beyond an upper range. INT2000 showed a lower-risk range between approximately 22 and 64, followed by a transition towards higher risk above 64. Skyline Articulation exhibited a similar pattern, indicating that greater visual complexity or urban intensity may not always correspond to more supportive environments. Distance to middle schools showed a comparable nonlinear pattern, although its interpretation was reversed because shorter distances indicate greater accessibility: distances of approximately 183–1,646 m were associated with lower risk, whereas values beyond about 1,686 m showed higher-risk responses.

Multi-stage patterns involved repeated transitions between lower- and higher-risk responses across the feature range. POI Entropy was mainly associated with lower risk between approximately 0.36 and 0.68, while lower values and a narrow high-value interval showed higher-risk responses, suggesting that moderate functional diversity may be more supportive. Dynamic Conflict Sources exhibited alternating response intervals, indicating sensitivity to local environmental intensity. UGS100 also showed multiple transitions before becoming consistently associated with lower risk above approximately 50. Compared with UGS300, this scale-dependent pattern suggests that broader green-space availability reflects overall environmental sufficiency, whereas local green-space provision is more sensitive to spatial configuration.

4.4. Spatial heterogeneity and social inequalities in preschooler health risk

4.4.1. Spatial heterogeneity in preschooler health risk

Based on the cross-domain lower- and higher-risk ranges, each street segment was represented by a multidimensional environmental response vector. Hierarchical clustering identified six health-oriented

street typologies with distinct environmental configurations (Figure 8). Neurodevelopmental outcomes and socioeconomic characteristics were not included in clustering but were subsequently used to interpret the resulting typologies. According to their environmental profiles and modelled neurodevelopmental risk levels, these typologies were further grouped into three broader categories: lower-risk, transitional-risk and higher-risk streets.

Spatially, the typologies exhibited a clear centre-periphery gradient. Lower-risk streets were mainly concentrated in the urban core, whereas higher-risk streets occurred more frequently in peripheral areas (Figure 8a). The **lower-risk category**, represented by High-Quality Core Streets (HCS), was characterised by favourable combinations of greenery, visual quality and pedestrian-oriented conditions, and showed the highest neurodevelopmental scores across domains. The **transitional-risk category** included streets with mixed environmental advantages and constraints. These streets generally benefited from service accessibility or functional concentration but showed deficiencies in greenery, streetscape quality or pedestrian safety, suggesting opportunities for targeted environmental improvement. In contrast, **higher-risk streets**, including Transport-Oriented Peripheral Corridors (TPC) and Constrained Edge Streets (CES), exhibited multiple disadvantages in accessibility, environmental quality and child-oriented street conditions, corresponding to lower neurodevelopmental scores.

Beyond environmental differences, socioeconomic patterns also varied across typologies (Figure 8c). Higher-risk streets were more frequently associated with lower parental educational attainment and household income, whereas lower-risk streets showed more favourable socioeconomic profiles. This alignment between environmental disadvantage and socioeconomic differentiation suggests that preschooler health-supportive environments are unevenly distributed across urban space, highlighting the need for place-based and equity-oriented planning interventions.

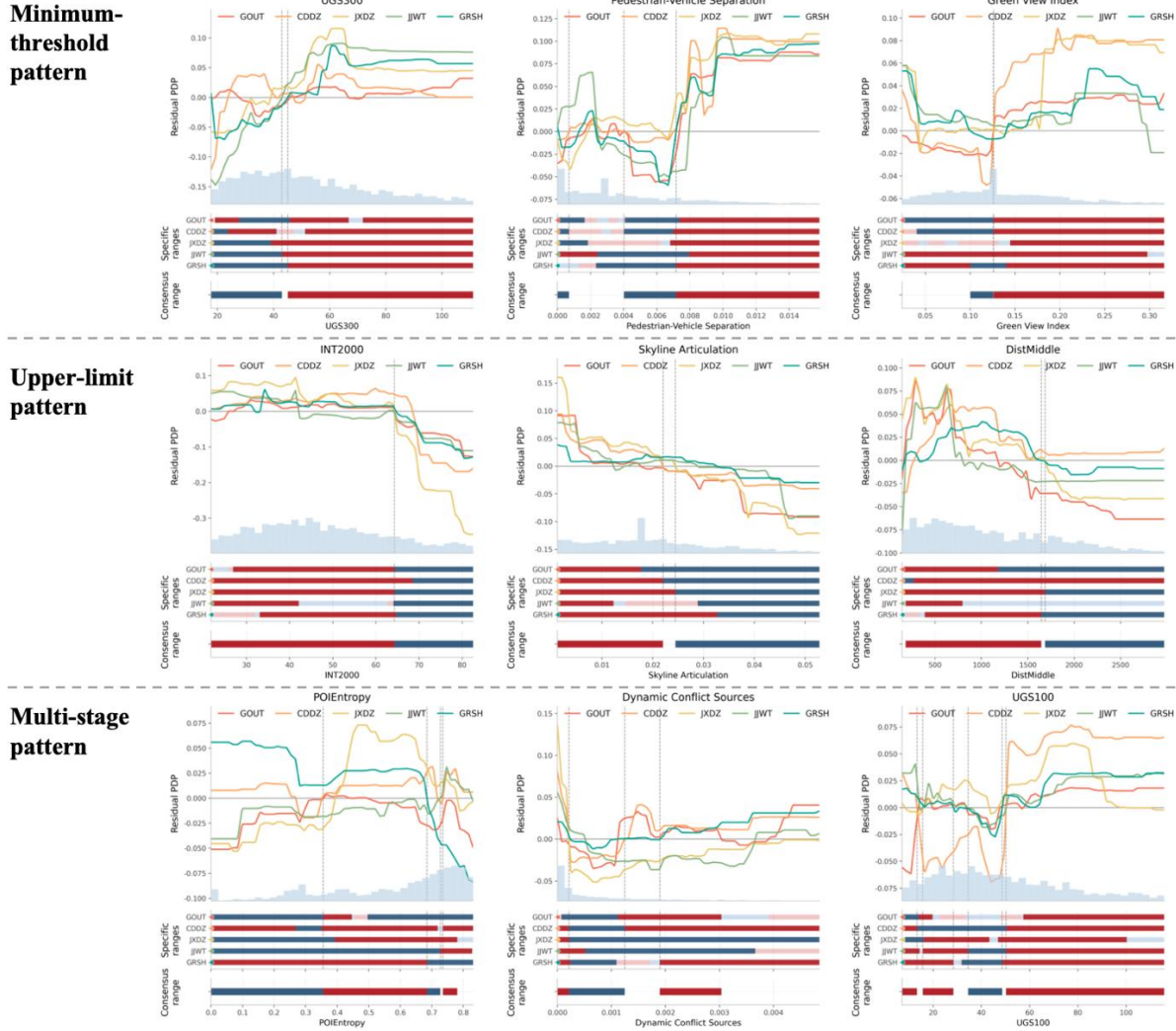


Figure 7 Nonlinear partial dependence profiles of selected built-environment features across preschooler neurodevelopmental domains.

4.4.2. Socio-spatial inequity to preschooler health risk

To examine whether health-oriented street environments were unevenly distributed across socioeconomic contexts, street segments were grouped into three tertiles according to aggregated maternal education levels, with Q1 representing the lowest and Q3 the highest socioeconomic context. The composition of street typologies across these strata was then compared to identify potential socioeconomic differentiation in environmental conditions.

A clear socioeconomic gradient was observed across the three groups (Figure 9). From Q3 to Q1, the proportion of lower-risk street environments progressively declined, whereas higher-risk and environmentally constrained typologies became increasingly prevalent (Figure 9a-b). Streets in higher-education contexts were more frequently associated with High-Quality Core Streets (HCS) and other relatively supportive environmental configurations, while lower-education contexts showed

greater representation of Constrained Edge Streets (CES) and Transport-Oriented Peripheral Corridors (TPC).

The environmental profiles further revealed systematic differences across socioeconomic strata (Figure 9c). Higher-education streets generally combined stronger service accessibility, functional diversity, greenery provision and favourable streetscape qualities. In contrast, lower-education streets exhibited weaker access to urban resources and greater exposure to environmental constraints, including poorer pedestrian conditions and higher traffic-related risks. Middle-education streets showed intermediate environmental characteristics, reflecting a transitional pattern between the two extremes.

These findings suggest that socioeconomic differentiation is reflected not only in the distribution of health-oriented street typologies but also in the underlying environmental configurations supporting children's everyday experiences. The concentration of less supportive street environments in lower socioeconomic contexts indicates a compound socio-

spatial disadvantage, where limited environmental resources and increased exposure to constraints coexist.

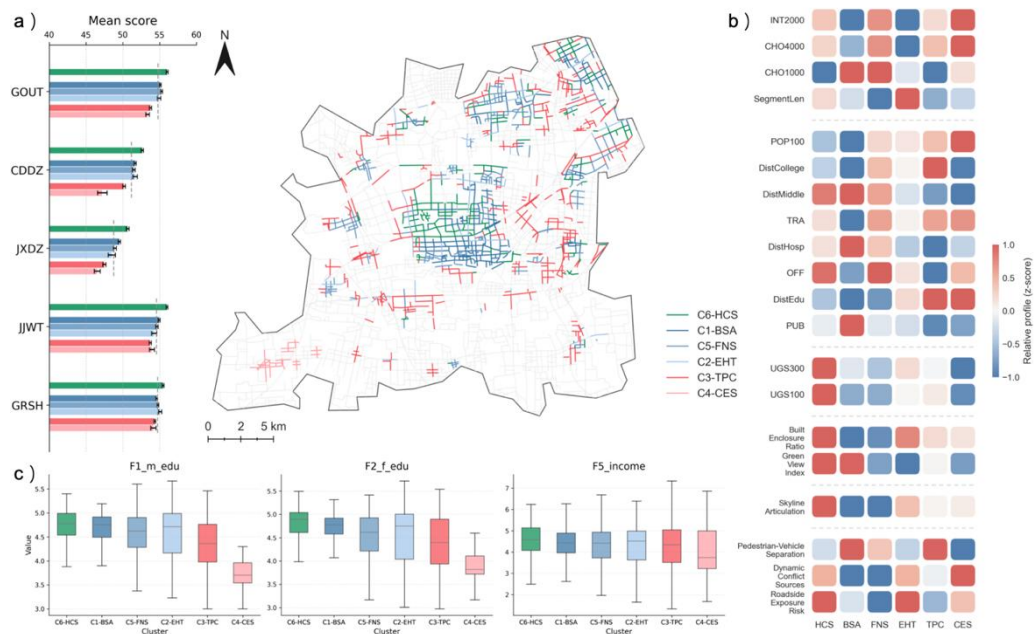


Figure 8 **Spatial, environmental and socioeconomic profiles of six health-oriented street typologies.** (a) Mean neurodevelopmental scores and spatial distribution of the six street typologies; (b) standardised place-based and human-focused built-environment profiles; and (c) distributions of maternal education, paternal education and household income across the six typologies. (HCS, High-Quality Core Streets; BSA, Balanced Service-Accessible Streets; FNS, Functional Node Streets; EHT, Enclosed High-Exposure Transition Streets; TPC, Transport-Oriented Peripheral Corridors; CES, Constrained Edge Streets.)

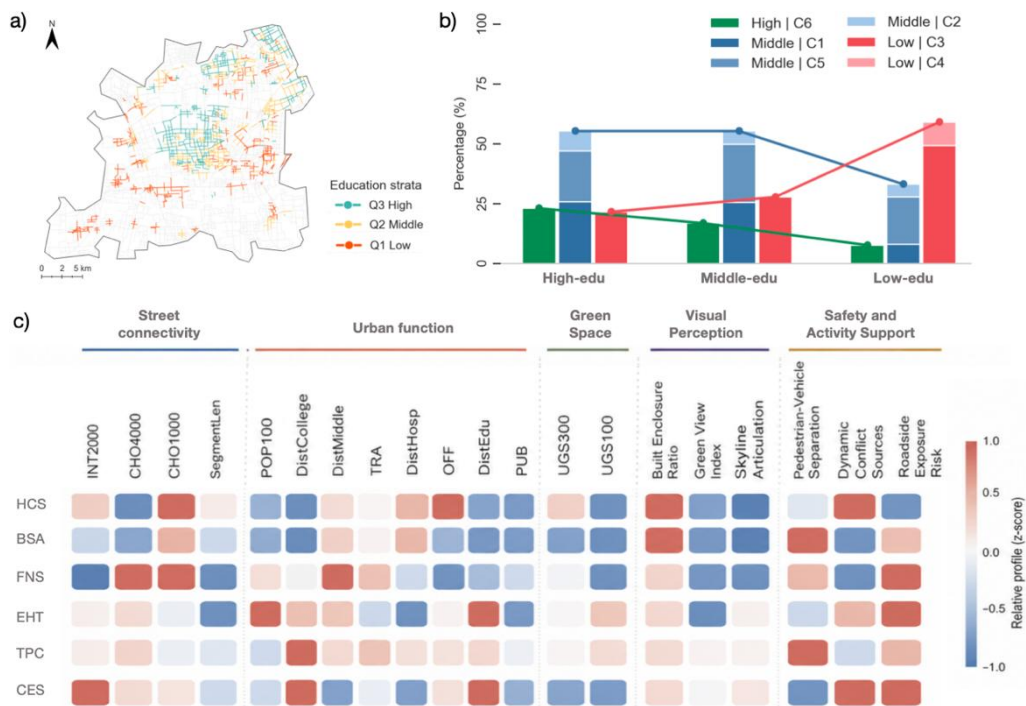


Figure 9 **Socio-spatial inequity to preschooler health risk and education-stratified built-environment association patterns.** (a) spatial distribution of maternal education strata. (b) composition of health-oriented street typologies within each education stratum. (c) profiles the environmental characteristics of each typology using standardized built-environment indicators across different built-environment dimensions

5. Discussion

5.1. Main findings

Street segments provided the most informative spatial representation among the evaluated alternatives, making associations between preschooler neurodevelopment and the built environment more spatially legible while also offering a fine-grained and operational unit for planning intervention. Compared with administrative neighbourhoods or predefined service areas (Villanueva et al., 2023), this representation preserved more local environmental variation and better reflected the network-based organisation of children's daily movement and exposure.

Residual modelling further showed that built-environment conditions contributed additional explanatory power beyond biodemographic characteristics and household socioeconomic conditions. Although the residual R^2 values were modest (0.049–0.067), they indicate that environmental conditions explained part of the neurodevelopmental variation that remained after accounting for individual and household factors. More importantly, the study translated model-derived environmental response ranges into health-oriented street typologies. Rather than identifying priority locations through single indicators or spatial clustering alone (Rui, 2023; Wu et al., 2024), these typologies captured multidimensional environmental configurations associated with different levels of neurodevelopmental risk and provided interpretable spatial units for differentiated planning interventions.

The results also revealed clear socioeconomic differentiation in health-oriented street environments. Streets in lower aggregated maternal education contexts were more likely to correspond to constrained and higher-risk typologies, suggesting that less supportive environments were unevenly distributed across socioeconomic groups. This finding aligns with previous evidence that child-friendly environmental resources and neighbourhood playability are often socioeconomically differentiated (Gemmell et al., 2026). Together, these findings highlight the importance of considering both environmental conditions and social context when designing interventions to support preschooler health.

5.2. Implications for policy

The framework supports spatially differentiated and equity-aware planning for child-supportive urban environments. First, street segments provide a practical spatial unit through which street-level patterns of environmental support and neurodevelopmental risk can be identified and translated into planning priorities. Second, the health-oriented street typology enables both the identification of priority locations and the selection of intervention strategies tailored to different environmental profiles. Lower-risk types should focus on maintaining existing environmental advantages and preventing decline during urban renewal. Transitional

types require targeted improvements addressing their specific constraints, such as limited service accessibility, pedestrian conditions or streetscape quality. Higher-risk streets, where multiple environmental disadvantages overlap, may require more comprehensive interventions combining service provision, traffic-conflict reduction, pedestrian safety improvement and public-space enhancement. Third, the socioeconomic differentiation of street-level risk profiles suggests that planning strategies should also consider social context. Streets in lower socioeconomic contexts may require targeted support to secure basic accessibility, safety and environmental quality, whereas relatively advantaged areas may benefit more from improving the diversity and quality of available opportunities.

5.3. Strengths and limitations

This study has several strengths. It focuses on preschoolers, an underexamined population during a critical period of cognitive, behavioural and socio-emotional development. By integrating standardised neurodevelopmental screening data with multidimensional and modifiable built-environment conditions at the street level, the study provides spatially explicit evidence that is directly relevant to urban planning intervention.

Several limitations should also be acknowledged. First, the study area is concentrated in Songjiang, a suburban district of Shanghai characterised by new-town development and urban-peripheral transition. Therefore, the identified environmental ranges and street typologies should not be directly generalised to other urban contexts, although the analytical framework is transferable. Broader and more diverse datasets are needed for metropolitan and cross-city validation. Second, the observational design captures residential environmental conditions rather than children's complete activity spaces, limiting causal interpretation. Future research incorporating longitudinal and mobility-based data could provide a more comprehensive understanding of children's everyday exposures. Third, although the study identifies socioeconomic differentiation in environmental risk profiles, further research is needed to directly examine how these inequalities evolve across different urban and social contexts.

6. Conclusion

This study addresses a key challenge in linking preschooler health evidence with urban planning practice: translating individual-level neurodevelopmental outcomes into actionable spatial knowledge. Street segments provided a fine-grained and operational representation for examining associations between preschooler neurodevelopment and the built environment, while built-environment conditions showed additional explanatory value beyond biodemographic and household socioeconomic factors. By integrating nonlinear environmental response patterns into health-

oriented street typologies, the framework moved beyond identifying individual environmental attributes to reveal distinct combinations of urban conditions and graded levels of neurodevelopmental risk across space. Furthermore, socioeconomic differentiation demonstrated that less supportive environmental profiles were unevenly distributed across social contexts. These findings provide a spatially explicit basis for identifying priority areas and developing differentiated interventions to create more supportive urban environments for preschoolers.

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